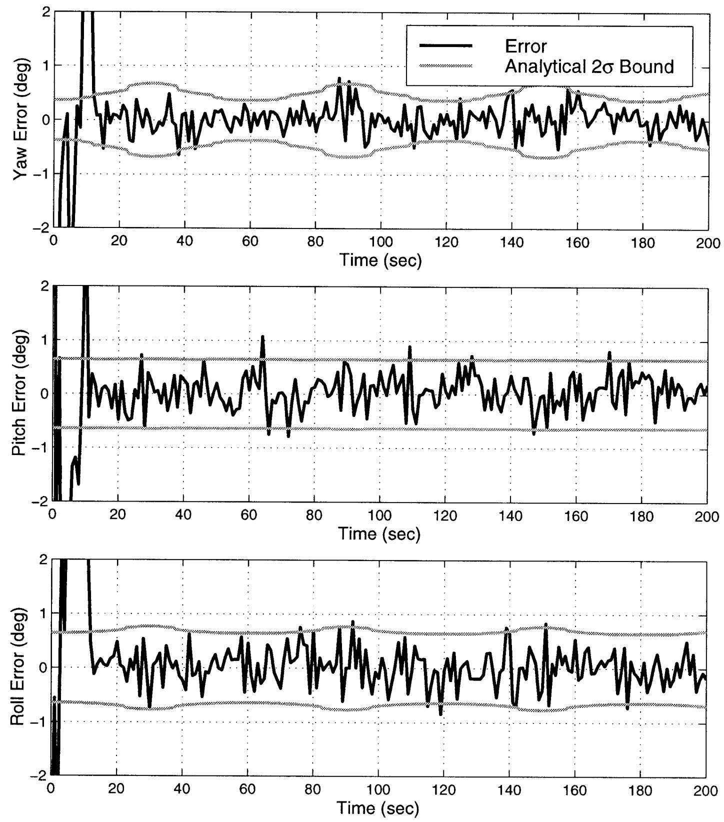

Fig. 5. Comparisons of attitude estimation errors against analytically calculated variances (case 2).

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Equations of Motion for a Two-Axes Gimbal System

Equations of motion for the two-axes yaw-pitch gimbal configuration are discussed on the assumption that the gimbals are rigid bodies and have no mass unbalance. The equations are derived and different terms are brought together into separate groups for the sake of clarity and to facilitate certain interpretations. Different kinds of disturbances and their elimination or reduction are discussed, as is the yaw gain dependence on the pitch angle. Inertia cross couplings are also given. The purpose is twofold: to simplify the picture of the two-axes gimbals and to further illustrate the properties of this configuration.

I. INTRODUCTION

Inertial stabilization of a sensor (such as IR, radar, or laser) is important in many aerospace applications,

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for example to satisfactorily carry out certain sensor functions, image processing, or target tracking. Basic stabilization is usually accomplished by suspending the sensor in a two-axes gimbal system, with the sensor on the inner gimbal. A two-axes rate gyro is usually placed on the inner gimbal as well, measuring the rotational rates in the two directions of interest. These gyro signals are utilized as feedback to torque motors acting on the gimbals, by that forming a control system, the purpose of which is to attenuate disturbances and to make the inner gimbal rotation follow a certain rate command (the regulator and servo case, respectively).

Even when suspended in a gimbal system, a sensor is exposed to certain disturbances. If these disturbances are successfully attenuated, it will be possible to keep the sensor optical axis nonrotating and consequently to maintain its direction in inertial space. Thus, stabilization is a question of attenuating disturbances. The underlying source of disturbances is in most cases a fuselage body motion, which influences the optical axis in a number of ways. Basically, we can think of these influences as having two main causes: the gimbal system as such, and imperfections. Even in a well-designed system some imperfections are inevitable, e.g. friction and mass unbalance, that will cause disturbance torques on the gimbals when the body is accelerating and rotating. Fundamental disturbances are caused also by basic properties of the gimbal system.

The two-axes yaw-pitch gimbal configuration is discussed here. This configuration is used in many systems and can be regarded as the archetype for other configurations, such as roll-pitch, mirror stabilization or four-axes. Furthermore, requirements on stabilization are increasing as a consequence of sensor resolution improvements (e.g. in the IR-case). Further treatment of this basic gimbal configuration should therefore be motivated.

The contributions of this work are as follows. The equations of motion are derived by the moment equation as well as the Lagrange equations, and it is shown that the equations can be given natural interpretations by suitable formulations. It is shown that gimbal disturbances are of two kinds: constraint disturbances, and inertia disturbances. The constraint disturbances can be regarded as the most fundamental. When a two-axes gimbal configuration has been chosen, the sensor is the object of some constraints that will cause disturbances which cannot be reduced by any choice of mechanical design parameters. It is pointed out that feedforward is a natural approach to reduce such disturbances. Also moments and products of inertia cause disturbances and it is shown that all inertia terms except one can be eliminated under certain inertia symmetry conditions on the mechanical design.

Furthermore, it is demonstrated that the loop gain in the yaw channel depends on the pitch angle in two ways and reduction of these dependencies is discussed. Inertia cross couplings between the channels are given and it is shown that these couplings can be eliminated by certain choices of inertia parameters. Input points for external torques are indicated in the model.

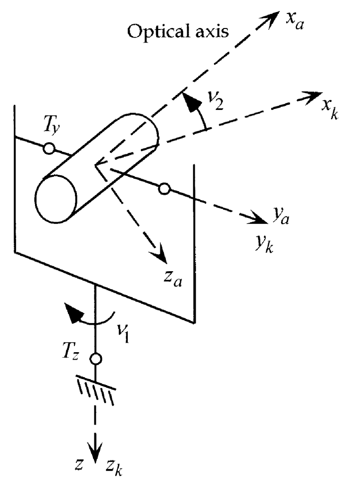

Fig. 1. Two-axes gimbal system.

Although implicit in earlier treatments [1–3], the approach and discussions here are believed to contribute further aspects. The purpose is twofold: to simplify the picture of the two-axes gimbals and to contribute additional insights into the properties of this configuration.

Notations, coordinate systems and transformations are introduced in Section II. The pitch and yaw channels are treated in Sections III and IV, respectively. Derivations and background materials from mechanics are collected in the Appendices.

II. REFERENCE FRAMES AND NOTATIONS

Consider a two-axes, yaw-pitch gimbal system as depicted in Fig. 1. In the figure the yaw and pitch gimbals are indicated. The sensor is placed on the pitch gimbal, or in fact *is* the pitch gimbal. The gimbals are regarded as rigid bodies. Three reference frames are introduced: a fuselage body-fixed frame B, a frame K fixed to the yaw gimbal, and a frame A fixed to the pitch gimbal. x_k, y_k, z_k and x_a, y_a, z_a are the yaw and pitch gimbal frame axes respectively. The x_a-axis coincides with the sensor optical axis. The z and z_k axes coincide and as usual in flight dynamics, the z axes are pointing "downwards." The center of rotation is in the frame origin, which is assumed to be the same point for the three frames. The angles of rotation ν_1 and ν_2 are defined in the following way.

1) The body-fixed frame B is carried into coincidence with the yaw gimbal frame K by a positive angle of rotation ν_1 about the z-axis.

2) The yaw gimbal frame K is carried into coincidence with the pitch gimbal frame A by a positive angle of rotation ν_2 about the y_k-axis.

Thus, B coincides with K for $\nu_1 = 0$. Associated with these rotations we have the following transformations [4]

$$L_{\mathrm{KB}} = \begin{bmatrix} \cos\nu_1 & \sin\nu_1 & 0 \\ -\sin\nu_1 & \cos\nu_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

$$L_{\mathrm{AK}} = \begin{bmatrix} \cos\nu_2 & 0 & -\sin\nu_2 \\ 0 & 1 & 0 \\ \sin\nu_2 & 0 & \cos\nu_2 \end{bmatrix}. \quad (2)$$

L_{KB} is the transformation from B to K, i.e., if a vector $\overline{v}_{\mathrm{B}}$ is expressed by its coordinates in frame B, then $L_{\mathrm{KB}}\overline{v}_{\mathrm{B}}$ gives the coordinates of the same vector in frame K. Similarly, L_{AK} is the transformation from K to A.

For the angular velocities of frames B, K, and A, respectively, we introduce

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix}; \quad \begin{bmatrix} p_k \\ q_k \\ r_k \end{bmatrix}; \quad \begin{bmatrix} p_a \\ q_a \\ r_a \end{bmatrix} \quad (3)$$

where p,q,r are the components in frame B of the inertial angular velocity vector of the frame B itself, and similarly for the other vectors. Following traditional practice in flight dynamics we use the notations p,q,r for the roll, pitch, and yaw components, respectively.

The inertia matrices of the gimbals are denoted by

$$\text{pitch gimbal}: \quad J_{\mathrm{A}} = \begin{bmatrix} J_{ax} & D_{xy} & D_{xz} \\ D_{xy} & J_{ay} & D_{yz} \\ D_{xz} & D_{yz} & J_{az} \end{bmatrix} \quad (4)$$

$$\text{yaw gimbal}: \quad J_{\mathrm{K}} = \begin{bmatrix} J_{kx} & d_{xy} & d_{xz} \\ d_{xy} & J_{ky} & d_{yz} \\ d_{xz} & d_{yz} & J_{kz} \end{bmatrix}. \quad (5)$$

Thus, moments of inertia are denoted by J and products of inertia by D and d (with appropriate indices). For the sign of the products of inertia, we follow the definition in [5]; consequently, no negative signs are explicitly used in the matrices. The mass center of a gimbal is supposed to be in the common center of rotation, that is, we assume that the gimbals have no mass unbalance.

Further, introduce (Fig. 1)

T_y = total external torque about the pitch gimbal y_a-axis

T_z = total external torque about the yaw gimbal z_k-axis.

Here the motor torque as well as external disturbance torques (e.g., friction torque) are included.

The angular velocities q_a and r_a are the outputs of the control system for stabilization, the purpose of which is to make it possible to keep $q_a = r_a = 0$ despite disturbances, and by that keep the sensor nonrotating in inertial space. q_a and r_a can be measured by a two-axes rate gyro on the pitch gimbal. Also body-fixed rate gyros can be used, from which q_a and r_a can be calculated utilizing the gimbal angels ν_1 and ν_2. In principle, there is no difference between these two ways of obtaining q_a and r_a. For convenience we can think of the gyro as placed on the gimbal.

III. PITCH CHANNEL

The basic equation of motion for the pitch gimbal is directly obtained if we think of this gimbal as a rigid body in its own right and consider the fact that there is no unbalance. Then we have

$$\begin{aligned} J_{ay}\dot{q}_a = T_y + (J_{az} - J_{ax})p_a r_a + D_{xz}(p_a^2 - r_a^2) \\ - D_{yz}(\dot{r}_a - p_a q_a) - D_{xy}(\dot{p}_a + q_a r_a). \end{aligned} \quad (6)$$

This is one component of the general equations of motion for a rigid body (Appendix A). It is a differential equation for the pitch angular velocity q_a, where external torque T_y, gimbal angular rates p_a,q_a,r_a and inertia parameters are included.

By using the transformations (36) and (37), (6) can be formulated in different ways, e.g. to express the influences of the body or the yaw gimbal. The equation will then be rather involved. However, to see the main properties of the pitch gimbal motion, the basic equation (6) is sufficient and suitable.

T_y is the sum of the motor torque and external imperfection disturbance torques. Clearly then, from stabilization point of view, the "inertia terms" on the right represent unwanted disturbances. They will enter the control system in the same point as an external torque; consequently, they can be regarded as torque disturbances.

From (6) we can see the effects of certain approximations and how inertia disturbances can be eliminated. Suppose for example that the products of inertia can be neglected

$$D_{xy} = D_{xz} = D_{yz} = 0. \quad (7)$$

Then (6) becomes one component of the well-known Euler equations of motion [5]. Assuming further that

$$J_{ax} = J_{az} \quad (8)$$

(6) becomes

$$J_{ay}\dot{q}_a = T_y. \quad (9)$$

If we disregard external disturbance torques, T_y represents only the motor torque. Then, (9) is the desired equation of motion, in the sense that no disturbances influence the motion about the pitch axis. No matter what body or yaw gimbal motions we have, the pitch gimbal angular velocity q_a is not affected.

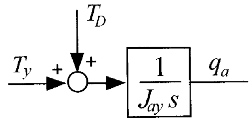

Fig. 2. Pitch channel.

The sensor is then isolated from the body, as far as pitch motions are concerned, and this is the ultimate goal for the pitch stabilization.

Even if a control system is used for stabilization, we should of course try to eliminate disturbances, that is, we should try to satisfy (7) and (8). This is a reasonable requirement on the system. Equations (7) and (8) are valid if the mass distribution is symmetrical with respect to the frame axes, which is a natural property of a sensor. The control system will then be used to attenuate external disturbances.

A. Block Diagram

The pitch equation (6) can be represented by the block diagram in Fig. 2, which we can also think of as the pitch channel. The disturbance term T_D has been introduced for the inertia terms, i.e.,

$$\begin{aligned} T_D = {} & (J_{az} - J_{ax}) p_a r_a + D_{xz}(p_a^2 - r_a^2) \\ & - D_{yz}(\dot{r}_a - p_a q_a) - D_{xy}(\dot{p}_a + q_a r_a). \end{aligned} \tag{10}$$

Thus, for the pitch channel we have a simple model: the motor torque and the disturbance torques are inputs to an integrator which includes the moment of inertia J_{ay}, and the output is the pitch angular velocity q_a. The integrator is the process to be controlled by feedback from q_a to the motor torque. $T_D = 0$ if (7) and (8) are satisfied.

From the control point of view, it is suitable to let T_y represent only the motor torque in the block diagram. External disturbance torques can then be included in T_D.

B. Cross Coupling

By using (37) the disturbance (10) can also be written

$$T_D = T_B + T_C \tag{11}$$

where

$$\begin{aligned} T_B = {} & -(D_{yz}\sin\nu_2 + D_{xy}\cos\nu_2)(\dot{p}_k + q_k r_k) \\ & + (D_{yz}\cos\nu_2 - D_{xy}\sin\nu_2) p_k q_k \\ & + [(J_{az} - J_{ax})\cos(2\nu_2) - 2D_{xz}\sin(2\nu_2)] p_k r_k \\ & + \tfrac{1}{2}[(J_{az} - J_{ax})\sin(2\nu_2) + 2D_{xz}\cos(2\nu_2)] p_k^2 \end{aligned} \tag{12}$$

$$\begin{aligned} T_C = {} & (D_{xy}\sin\nu_2 - D_{yz}\cos\nu_2)\dot{r}_k \\ & - \tfrac{1}{2}[(J_{az} - J_{ax})\sin(2\nu_2) + 2D_{xz}\cos(2\nu_2)] r_k^2. \end{aligned} \tag{13}$$

From this separation of T_D into two parts, it is clear that the yaw gimbal may influence the pitch gimbal irrespective of body motions. Suppose that the body is nonrotating, $p = q = r = 0$. Then from (36) we have $p_k = q_k = 0$ and by that $T_B = 0$. However, r_k and consequently T_C is not necessarily zero for this case; yaw gimbal motions may influence the pitch gimbal via T_C even for a nonrotating body.

As we will see, there is a corresponding influence from the pitch gimbal on the yaw gimbal. From the control point of view, T_C is a cross coupling term, while T_B represents disturbances caused by body rotations. We note that $T_B = 0$ or $T_C = 0$ only if both (7) and (8) are satisfied.

IV. YAW CHANNEL

A. Yaw Gimbal Equation

As demonstrated in Appendix B, the following equation of motion can be derived for the yaw gimbal:

$$J_k \dot{r}_k = T_z + T_{d1} + T_{d2} + T_{d3} \tag{14}$$

where

$$J_k = J_{kz} + J_{ax}\sin^2\nu_2 + J_{az}\cos^2\nu_2 - D_{xz}\sin(2\nu_2) \tag{15}$$

$$\begin{aligned} T_{d1} = {} & [J_{kx} + J_{ax}\cos^2\nu_2 + J_{az}\sin^2\nu_2 \\ & + D_{xz}\sin(2\nu_2) - (J_{ky} + J_{ay})] p_k q_k \end{aligned} \tag{16}$$

$$\begin{aligned} T_{d2} = {} & -[d_{xz} + (J_{az} - J_{ax})\sin\nu_2\cos\nu_2 + D_{xz}\cos(2\nu_2)] \\ & \times (\dot{p}_k - q_k r_k) \\ & - (d_{yz} + D_{yz}\cos\nu_2 - D_{xy}\sin\nu_2)(\dot{q}_k + p_k r_k) \\ & - (d_{xy} + D_{xy}\cos\nu_2 + D_{yz}\sin\nu_2)(p_k^2 - q_k^2) \end{aligned} \tag{17}$$

$$\begin{aligned} T_{d3} = {} & \ddot{\nu}_2(D_{xy}\sin\nu_2 - D_{yz}\cos\nu_2) \\ & + \dot{\nu}_2[(J_{ax} - J_{az})(p_k\cos(2\nu_2) - r_k\sin(2\nu_2)) \\ & \quad + 2D_{xz}(p_k\sin(2\nu_2) + r_k\cos(2\nu_2)) \\ & \quad + (D_{yz}\sin\nu_2 + D_{xy}\cos\nu_2)(q_a + q_k) - J_{ay}p_k]. \end{aligned} \tag{18}$$

Equation (14) is a differential equation for the yaw gimbal motion r_k, where T_z is the total external torque about the yaw gimbal z_k-axis. T_{d1}, T_{d2}, and T_{d3} represent different gimbal inertia disturbances.

This equation of motion can be formulated in different ways and for different purposes by using the transformations in Appendix A. The body rotations could for example be taken into account. However, the equations tend to be rather complicated then. The exposition (14)–(18) is used to give a simplified picture of the yaw gimbal motion. In this formulation, the terms have been collected in different groups in such a way, that a natural interpretation is possible from the relations in Appendix A.

First, we note from (43c) that J_k is the total moment of inertia of the gimbal system about the yaw gimbal z_k-axis, if the pitch gimbal has been rotated an angle ν_2. Thus, even if the pitch gimbal is movable and the gimbal angle ν_2 can change its value, J_k is the instantaneous moment of inertia about the z_k-axis. Naturally then, J_k is the moment of inertia factor in the equation of motion (14).

The disturbances T_{d1} and T_{d2} can also be interpreted in a natural way. Suppose that the pitch gimbal has been rotated an angle ν_2 and is fixed in this position; hence $\dot{\nu}_2 \equiv 0$ and by that $T_{d3} \equiv 0$. Then, from (41) and (43) it is clear that T_{d1} and T_{d2} are the inertia disturbances in the rigid body equation of motion for the total gimbal system about the z_k-axis. T_{d1} is the term obtained from the difference between two moments of inertia and T_{d2} is the sum of the three terms obtained from the products of inertia. That the disturbances can be formulated in this way is not surprising, since the gimbal system actually is a rigid body when ν_2 is fixed. Compared with the pitch gimbal, the only complication here is the fact that moments and products of inertia depend on ν_2.

If ν_2 is not fixed and $\dot{\nu}_2 \neq 0$, the total gimbal system is no longer a rigid body. However, T_{d1} and T_{d2} can still be regarded as rigid body inertia disturbances, now obtained for the instantaneous value of ν_2. An additional disturbance T_{d3} is obtained, though, because of the pitch gimbal motion relative to the yaw gimbal, as expressed by $\dot{\nu}_2$ and $\ddot{\nu}_2$. We can think of T_{d1} and T_{d2} as the "rigid body components" of the equation of motion and T_{d3} as the "deformation component."

Thus, although quite a number of inertia and trigonometric terms are involved, (14) has a reasonable and simple interpretation. This will be even more apparent, if we consider the significance of certain inertia conditions for disturbance elimination. Suppose that

$$D_{xy} = D_{xz} = D_{yz} = 0 \tag{19}$$

$$J_{ax} = J_{az}. \tag{20}$$

These are the inertia properties, that we should try to attain for the pitch gimbal. On these assumptions most of the disturbances in (14)–(18) will disappear. Thus, the conditions (19) and (20) are valuable for the yaw gimbal motion as well. Furthermore, if the products of inertia of the yaw gimbal can also be neglected, that is if

$$d_{xy} = d_{xz} = d_{yz} = 0 \tag{21}$$

which is a natural symmetry property, we get

$$J_k = J_{kz} + J_{az} \tag{22}$$

$$T_{d1} = [J_{kx} + J_{ax} - (J_{ky} + J_{ay})]p_k q_k \tag{23}$$

$$T_{d2} = 0 \tag{24}$$

$$T_{d3} = -J_{ay}\dot{\nu}_2 p_k. \tag{25}$$

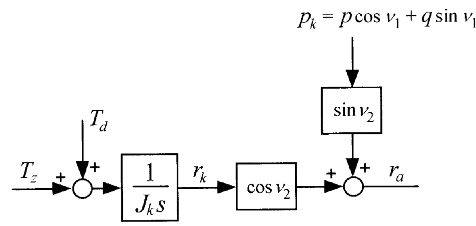

Fig. 3. Yaw channel.

Only a few disturbances are now left, and it is clearly seen that T_{d1} is in fact determined by the difference between the two moments of inertia associated with the x_k- and y_k-axes. Using that $\dot{\nu}_2 = q_a - q_k$ we get

$$T_{d1} + T_{d2} + T_{d3} = (J_{kx} + J_{ax} - J_{ky})p_k q_k - J_{ay} p_k q_a. \tag{26}$$

We still have a possibility here to eliminate disturbances by requiring

$$J_{kx} + J_{ax} = J_{ky}. \tag{27}$$

If this could be fulfilled, there is only one inertia disturbance $J_{ay}p_k q_a$ left in the whole gimbal system. This term cannot be eliminated, though, by any inertia relationship. Obviously, this is the limit of what can be achieved, if we try to reduce disturbances by inertia requirements.

B. Block Diagram

We have obtained an equation of motion for the yaw gimbal, that is for r_k. However, the output in the yaw *channel* is the angular velocity r_a, the yaw angular velocity of the pitch gimbal (i.e., of the sensor). This is the variable that we want to control, and this can be accomplished from the yaw torque motor, since r_k is related to r_a. From (37) we have

$$r_a = p_k \sin\nu_2 + r_k \cos\nu_2. \tag{28}$$

Thus, r_a can be influenced by r_k but also by p_k which is an undesired effect. From (36) we have

$$p_k = p\cos\nu_1 + q\sin\nu_1 \tag{29}$$

and we see that p_k is related to the body motions p and q. For the yaw channel we now get the block diagram in Fig. 3. The notation T_d has been introduced for the inertia disturbances, i.e.,

$$T_d = T_{d1} + T_{d2} + T_{d3}. \tag{30}$$

T_z and T_d are the inputs to an integrator, where the moment of inertia J_k is included, which gives the yaw gimbal angular velocity r_k. If we let T_z represent only the motor torque, all other external torques can be regarded as included in T_d. The output r_a is the variable to be controlled by feedback to the motor torque. p_k is a disturbance acting directly on the output.

C. Properties of Yaw Channel

Compared with the pitch channel, we have some important differences, which restrict the achievable performance of the two-axes gimbal configuration.

First we note that $\cos\nu_2$ is a factor in the loop gain, that is, the gain depends on the pitch gimbal angle ν_2. Such a loop gain dependence may be troublesome, since it means difficulties to obtain the required closed loop bandwidth for different pitch angles. A natural although approximate way to handle this is to divide by $\cos\nu_2$, i.e., to let the controller have a factor $1/\cos\nu_2$. If ν_2 changes slowly compared with the time constants involved, which is normally the case, the dependence will then be almost eliminated.

However, when $\nu_2 \to \pi/2$ this way to compensate for $\cos\nu_2$ will increase the demands on the maximum motor torque and, of course, it will not work for $\nu_2 = \pi/2$, the so-called singular point of the gimbal system. We have run into a basic restriction in a two-axes system: the singular point and values close to this point should be avoided. In the block diagram this is represented by the factor $\cos\nu_2$.

We note from (15) that we also have a ν_2-dependence in J_k and by that another ν_2-dependence in the loop gain. However, by the inertia conditions (19) and (20) this dependence is eliminated. This is another advantage of these conditions: besides the removal of disturbances, also a loop gain variation is eliminated.

Two disturbances influence the yaw channel: inertia disturbances T_d, and a constraint disturbance $p_k \sin\nu_2$. It should be emphasized that these are different kinds of disturbances. T_d is the sum of a number of inertia terms, that is, disturbances where moments and products of inertia are involved. As we have seen, all these terms except one can be eliminated under certain inertia conditions. Most of these conditions are fulfilled if the gimbals have a certain mass symmetry. This also means, that we can think of most of the disturbances in T_d as caused by an unsymmetrical mass distribution in the gimbal system.

$p_k \sin\nu_2$ on the other hand cannot be eliminated in this way, since no inertia is involved here. We can think of this disturbance as obtained from the constraints exerted by the gimbal system. We note that it has no correspondence in the pitch channel. It is an important and troublesome disturbance, generated by the body rotations p and q and influencing the output directly. Attenuation of such a direct output disturbance may be difficult and calls for a high performance servo system, i.e., a system with high bandwidth.

In many applications the body motions p and q are measured by rate gyros in the system, that is, they are known quantities. Also the gimbal angles are usually measured. The disturbance $p_k \sin\nu_2$ is then known and can be used for feedforward control. As we know, in practice feedforward cannot be perfectly realized, but also an approximation is useful. The influence of the disturbance can then be reduced, which means a corresponding reduction of the requirements on the closed loop.

D. Cross Coupling

Using in (18) that $\dot{\nu}_2 = q_a - q_k$ the disturbance T_d can be written as

$$T_d = T_b + T_c \tag{31}$$

where

$$\begin{aligned} T_b = {} & T_{d1} + T_{d2} + [(J_{ax} - J_{az})\cos(2\nu_2) + 2D_{xz}\sin(2\nu_2) - J_{ay}]\dot{\nu}_2 p_k \\ & + [(J_{ax} - J_{az})\sin(2\nu_2) - 2D_{xz}\cos(2\nu_2)]q_k r_k \\ & + (D_{yz}\cos\nu_2 - D_{xy}\sin\nu_2)\dot{q}_k - (D_{yz}\sin\nu_2 + D_{xy}\cos\nu_2)q_k^2 \end{aligned} \tag{32}$$

$$\begin{aligned} T_c = {} & (D_{xy}\sin\nu_2 - D_{yz}\cos\nu_2)\dot{q}_a + (D_{xy}\cos\nu_2 + D_{yz}\sin\nu_2)q_a^2 \\ & + [(J_{az} - J_{ax})\sin(2\nu_2) + 2D_{xz}\cos(2\nu_2)]q_a r_k. \end{aligned} \tag{33}$$

Here we have a situation similar to the pitch gimbal. The disturbance T_d can be divided into two parts, where one part represents a cross coupling. For a nonrotating body $p = q = r = 0$, we have $p_k = q_k = 0$ and by that $T_b = 0$. However, irrespective of the body motion, we also have a cross coupling term T_c influencing the yaw gimbal from pitch gimbal motions. T_c is independent of the yaw gimbal inertia parameters and we have $T_c = 0$ if the pitch gimbal conditions (19) and (20) are satisfied.

V. CONCLUSION

The two-axes yaw-pitch gimbal configuration is discussed and the purpose is twofold: to simplify our conception of the two-axes gimbals and to further illustrate the properties of this configuration.

The equations of motion are derived, by the moment equation as well as the Lagrange equations, and it is demonstrated that the equations can be formulated in such a way, that they can be given natural interpretations. It is shown that the disturbances are of two kinds: inertia and constraint disturbances. How to eliminate or reduce these disturbances is discussed. All inertia disturbances except one can be eliminated by certain inertia symmetry conditions and feedforward can be used to reduce the constraint disturbance.

It is shown that the loop gain in the yaw channel depends on the pitch angle in two ways and reduction or elimination of these dependencies is discussed. Inertia cross couplings between the channels are given and it is shown that certain choices of inertia parameters will eliminate these couplings.

APPENDIX A

A. Angular Velocity and Rigid Body Relationships

The orientation of two reference frames relative each other can be given by Euler angles, i.e., by three consecutive rotations ψ, θ, ϕ about the z, y, x axes respectively of one system to carry it into coincidence with the other. The order is essential: a rotation ψ about the z-axis is followed by a rotation θ about the y-axis and finally by a rotation ϕ about the x-axis. This is the definition of Euler angles currently used in flight dynamics.

For two frames F and G the following relation can then be derived [4]

$$\overline{\omega}^{\mathrm{G}} - \overline{\omega}^{\mathrm{F}} = \begin{bmatrix} \dot{\phi} - \dot{\psi}\sin\theta \\ \dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi \\ \dot{\psi}\cos\theta\cos\phi - \dot{\theta}\sin\phi \end{bmatrix}_{\mathrm{G}} \tag{34}$$

$\overline{\omega}^{\mathrm{F}}$ and $\overline{\omega}^{\mathrm{G}}$ are the inertial angular velocity vectors of frame F and frame G, respectively, and frame F is carried into frame G by the Euler angles ψ, θ, ϕ. The difference between these angular velocities is a vector, given by the right-hand side, where the components are expressed in frame G.

Between our body-fixed frame B and yaw gimbal frame K we have only one rotation ν_1 about the z-axis. With F as B and G as K, then $\theta = \phi = 0$ and $\psi = \nu_1$ in (34). Using the angular velocities (3) we get from (34)

$$\begin{bmatrix} p_k \\ q_k \\ r_k \end{bmatrix} - L_{\mathrm{KB}} \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \dot{\nu}_1 \end{bmatrix}. \tag{35}$$

The vectors are expressed in system K. Inserting transformation (1) gives

$$\begin{aligned} p_k &= p\cos\nu_1 + q\sin\nu_1 \\ q_k &= -p\sin\nu_1 + q\cos\nu_1 \\ r_k &= r + \dot{\nu}_1. \end{aligned} \tag{36}$$

Similarly, between the yaw gimbal K and the pitch gimbal A we have $\psi = \phi = 0$, $\theta = \nu_2$ in (34). Using transformation (2) we get for this case

$$\begin{aligned} p_a &= p_k\cos\nu_2 - r_k\sin\nu_2 \\ q_a &= q_k + \dot{\nu}_2 \\ r_a &= p_k\sin\nu_2 + r_k\cos\nu_2. \end{aligned} \tag{37}$$

The gimbal angles and the different angular velocities are usually functions of time, caused by certain "inputs" to the equations of motion, such as body motions and motor torques. However, at every moment of time (36) and (37) are necessarily valid, and could be viewed as basic properties of the gimbal configuration. We can also think of (36) and (37) as mathematical expressions for the constraints exerted by the gimbal configuration.

Consider a rigid body and a reference frame x, y, z fixed to the body. Let

$$J = \begin{bmatrix} J_x & D_{xy} & D_{xz} \\ D_{xy} & J_y & D_{yz} \\ D_{xz} & D_{yz} & J_z \end{bmatrix} \tag{38}$$

be the inertia matrix and denote by

$$\overline{\omega} = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \tag{39}$$

the inertial angular velocity of the body expressed in the body-fixed frame. Then, in this system the angular momentum is $\overline{H} = J\overline{\omega}$. The moment equation for a rotating frame is $\overline{T} = d\overline{H}/dt + \overline{\omega} \times \overline{H}$ where $\overline{T}$ is the external torque. From these relationships the rigid body equations of motion can be obtained, and the y and z components are

$$\begin{aligned} T_y &= J_y\dot{q} + (J_x - J_z)pr + D_{xz}(r^2 - p^2) \\ &\quad + D_{xy}(\dot{p} + qr) + D_{yz}(\dot{r} - pq) \end{aligned} \tag{40}$$

$$\begin{aligned} T_z &= J_z\dot{r} + (J_y - J_x)pq + D_{xy}(p^2 - q^2) \\ &\quad + D_{yz}(\dot{q} + pr) + D_{xz}(\dot{p} - qr). \end{aligned} \tag{41}$$

We also ask for the inertia matrix in K of the total gimbal system. The pitch gimbal inertia matrix J_{A} is transformed to the system K by the transformation $L_{\mathrm{AK}}^{\mathrm{T}} J_{\mathrm{A}} L_{\mathrm{AK}}$ [4]. Using the notations (4) and (5), we obtain the inertia matrix of the total gimbal system with respect to K:

$$J_{\mathrm{K}} + L_{\mathrm{AK}}^{\mathrm{T}} J_{\mathrm{A}} L_{\mathrm{AK}} = \begin{bmatrix} J'_x & D'_{xy} & D'_{xz} \\ D'_{xy} & J'_y & D'_{yz} \\ D'_{xz} & D'_{yz} & J'_z \end{bmatrix} \tag{42}$$

where

$$J'_x = J_{kx} + J_{ax}\cos^2\nu_2 + J_{az}\sin^2\nu_2 + D_{xz}\sin(2\nu_2) \tag{43a}$$

$$J'_y = J_{ky} + J_{ay} \tag{43b}$$

$$J'_z = J_{kz} + J_{ax}\sin^2\nu_2 + J_{az}\cos^2\nu_2 - D_{xz}\sin(2\nu_2) \tag{43c}$$

$$D'_{xy} = d_{xy} + D_{xy}\cos\nu_2 + D_{yz}\sin\nu_2 \tag{43d}$$

$$D'_{xz} = d_{xz} + (J_{az} - J_{ax})\sin\nu_2\cos\nu_2 + D_{xz}\cos(2\nu_2) \tag{43e}$$

$$D'_{yz} = d_{yz} + D_{yz}\cos\nu_2 - D_{xy}\sin\nu_2. \tag{43f}$$

APPENDIX B

A. Derivation of Equations of Motion from Moment Equation

The Pitch Gimbal: The equation of motion for the pitch gimbal has in fact already been given in (40).

When this equation is applied to the pitch gimbal, the equation of motion (6) is immediately obtained.

The Yaw Gimbal: The angular momentum of the total gimbal system is the sum of the angular momentum of the pitch and yaw gimbals. The angular momentum for a gimbal is given by $J\overline{\omega}$, where J is the inertia matrix and $\overline{\omega}$ the angular velocity. With the inertia matrices (4) and (5) and the velocities (3), we get

$$\overline{H} = \begin{bmatrix} H_x \\ H_y \\ H_z \end{bmatrix} = \begin{bmatrix} J_{kx}p_k + d_{xy}q_k + d_{xz}r_k \\ d_{xy}p_k + J_{ky}q_k + d_{yz}r_k \\ d_{xz}p_k + d_{yz}q_k + J_{kz}r_k \end{bmatrix} + L_{\mathrm{AK}}^{\mathrm{T}} \begin{bmatrix} J_{ax}p_a + D_{xy}q_a + D_{xz}r_a \\ D_{xy}p_a + J_{ay}q_a + D_{yz}r_a \\ D_{xz}p_a + D_{yz}q_a + J_{az}r_a \end{bmatrix}. \quad (44)$$

The second term on the right-hand side is the angular momentum of the pitch gimbal transformed to the yaw gimbal frame K. Then, $\overline{H}$ is the total angular momentum expressed in frame K. The components of $\overline{H}$ in this frame are also introduced. We get

$$H_z = d_{xz}p_k + d_{yz}q_k + J_{kz}r_k - (J_{ax}p_a + D_{xy}q_a + D_{xz}r_a)\sin\nu_2 + (D_{xz}p_a + D_{yz}q_a + J_{az}r_a)\cos\nu_2. \quad (45)$$

With $\overline{\omega} = [p_k, q_k, r_k]^{\mathrm{T}}$, the angular velocity of frame K, we have the z-component

$$\begin{aligned}(\overline{\omega}\times\overline{H})_z &= p_kH_y - q_kH_x \\ &= p_k(d_{xy}p_k + J_{ky}q_k + d_{yz}r_k + D_{xy}p_a + J_{ay}q_a + D_{yz}r_a) \\ &\quad - q_k(J_{kx}p_k + d_{xy}q_k + d_{xz}r_k) \\ &\quad - q_k(J_{ax}p_a + D_{xy}q_a + D_{xz}r_a)\cos\nu_2 \\ &\quad - q_k(D_{xz}p_a + D_{yz}q_a + J_{az}r_a)\sin\nu_2. \end{aligned} \quad (46)$$

The equation of motion for the yaw gimbal can now be obtained. It is the z-component of the moment equation $\overline{T} = d\overline{H}/dt + \overline{\omega}\times\overline{H}$ applied to frame K (the moment equation is valid also for a deformable body). After some algebraic calculations, the details of which are omitted, where (45), (46) and (37) are used, this equation can be formulated in the following way:

$$J_k\dot{r}_k = T_z + T_{d1} + T_{d2} + T_{d3} \quad (47)$$

where

$$J_k = J_{kz} + J_{ax}\sin^2\nu_2 + J_{az}\cos^2\nu_2 - D_{xz}\sin(2\nu_2) \quad (48)$$

$$T_{d1} = [J_{kx} + J_{ax}\cos^2\nu_2 + J_{az}\sin^2\nu_2 + D_{xz}\sin(2\nu_2) - (J_{ky} + J_{ay})]p_kq_k \quad (49)$$

$$\begin{aligned}T_{d2} &= -[d_{xz} + (J_{az} - J_{ax})\sin\nu_2\cos\nu_2 + D_{xz}\cos(2\nu_2)] \\ &\quad \times(\dot{p}_k - q_kr_k) \\ &\quad - (d_{yz} + D_{yz}\cos\nu_2 - D_{xy}\sin\nu_2)(\dot{q}_k + p_kr_k) \\ &\quad - (d_{xy} + D_{xy}\cos\nu_2 + D_{yz}\sin\nu_2)(p_k^2 - q_k^2) \end{aligned} \quad (50)$$

$$\begin{aligned}T_{d3} &= \ddot{\nu}_2(D_{xy}\sin\nu_2 - D_{yz}\cos\nu_2) \\ &\quad + \dot{\nu}_2[(J_{ax} - J_{az})(p_k\cos(2\nu_2) - r_k\sin(2\nu_2)) \\ &\quad + 2D_{xz}(p_k\sin(2\nu_2) + r_k\cos(2\nu_2)) \\ &\quad + (D_{yz}\sin\nu_2 + D_{xy}\cos\nu_2)(q_a + q_k) - J_{ay}p_k]. \end{aligned} \quad (51)$$

B. Derivation of Equations of Motion from Lagrange Equations

The orientation of the gimbal system in an inertial system is completely determined by four independent consecutive rotations ϕ, θ, ψ, ν_2 where ϕ, θ, ψ are three Euler rotations of the yaw gimbal and ν_2 is the earlier introduced pitch gimbal angle. Then we can take these angles of rotations as the generalized coordinates in the Lagrange equations. The order of rotation is essential. By taking the rotations in the order roll (ϕ), pitch (θ), yaw (ψ) followed by ν_2, the generalized "forces" corresponding to the coordinates ψ and ν_2 are the external torques T_z and T_y applied to the yaw and pitch gimbal, respectively.

The kinetic energy of a rotating body is given by the scalar product $\overline{\omega}\cdot\overline{H}/2$ [5]. Then, using (4) and (5) we get the kinetic energy for our gimbal system:

$$\begin{aligned}T &= \tfrac{1}{2}(J_{ax}p_a^2 + J_{ay}q_a^2 + J_{az}r_a^2) + D_{xy}p_aq_a \\ &\quad + D_{xz}p_ar_a + D_{yz}q_ar_a \\ &\quad + \tfrac{1}{2}(J_{kx}p_k^2 + J_{ky}q_k^2 + J_{kz}r_k^2) \\ &\quad + d_{xy}p_kq_k + d_{xz}p_kr_k + d_{yz}q_kr_k. \end{aligned} \quad (52)$$

The common notation T is used for kinetic energy, while T_y and T_z denote torques. A translation velocity contributes to the kinetic energy, but since we have assumed that the gimbal system has no mass unbalance, a translation motion has no influence on the rotational equations of motion. Then there is no need to take such velocity terms into account.

For the yaw gimbal angular velocities and the Euler angles and their time derivatives, the following relations can be derived

$$\begin{aligned}p_k &= \dot{\phi}\cos\theta\cos\psi + \dot{\theta}\sin\psi \\ q_k &= -\dot{\phi}\cos\theta\sin\psi + \dot{\theta}\cos\psi \\ r_k &= \dot{\phi}\sin\theta + \dot{\psi}. \end{aligned} \quad (53)$$

These are relationships of the same kind as (34), for the order of rotations defined here and starting the rotations from an inertial system.

Using (53) and (37) in (52) now gives the kinetic energy as a function of the generalized coordinates and their time derivatives. We can then formulate the Lagrange equations and by that obtain the equations of motion. This is demonstrated here for the yaw gimbal.

The Lagrange equation for ψ is

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\psi}}\right) - \frac{\partial T}{\partial \psi} = T_z \tag{54}$$

where T is the kinetic energy (52) and T_z is the torque about the yaw gimbal z_k-axis. From (53) we have

$$\begin{aligned}
\frac{\partial p_k}{\partial \dot{\psi}} &= 0 \qquad & \frac{\partial p_k}{\partial \psi} &= q_k \\
\frac{\partial q_k}{\partial \dot{\psi}} &= 0 \qquad & \frac{\partial q_k}{\partial \psi} &= -p_k \\
\frac{\partial r_k}{\partial \dot{\psi}} &= 1 \qquad & \frac{\partial r_k}{\partial \psi} &= 0
\end{aligned} \tag{55}$$

and from (37)

$$\begin{aligned}
\frac{\partial p_a}{\partial \dot{\psi}} &= -\sin\nu_2 \qquad & \frac{\partial p_a}{\partial \psi} &= q_k \cos\nu_2 \\
\frac{\partial q_a}{\partial \dot{\psi}} &= 0 \qquad & \frac{\partial q_a}{\partial \psi} &= -p_k \\
\frac{\partial r_a}{\partial \dot{\psi}} &= \cos\nu_2 \qquad & \frac{\partial r_a}{\partial \psi} &= q_k \sin\nu_2.
\end{aligned} \tag{56}$$

This gives

$$\begin{aligned}
\frac{\partial T}{\partial \dot{\psi}} &= d_{xz}p_k + d_{yz}q_k + J_{kz}r_k - (J_{ax}p_a + D_{xy}q_a + D_{xz}r_a)\sin\nu_2 \\
&\quad + (D_{xz}p_a + D_{yz}q_a + J_{az}r_a)\cos\nu_2
\end{aligned} \tag{57}$$

$$\begin{aligned}
\frac{\partial T}{\partial \psi} &= -p_k(d_{xy}p_k + J_{ky}q_k + d_{yz}r_k + D_{xy}p_a + J_{ay}q_a + D_{yz}r_a) \\
&\quad + q_k(J_{kx}p_k + d_{xy}q_k + d_{xz}r_k) \\
&\quad + q_k(J_{ax}p_a + D_{xy}q_a + D_{xz}r_a)\cos\nu_2 \\
&\quad + q_k(D_{xz}p_a + D_{yz}q_a + J_{az}r_a)\sin\nu_2.
\end{aligned} \tag{58}$$

A comparison with (45) shows that (57) is in fact H_z and from (46) it is seen that (58) is $-(\overline{\omega} \times \overline{H})_z$. Thus, it follows that the Lagrange equation (54) is equivalent to the z-component of the moment equation applied to system K as above. Consequently, the Lagrange equation gives the same equations of motion (47)–(51).

The pitch gimbal equation is obtained similarly.

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Time-Frequency Approaches to ISAR Imaging of Maneuvering Targets and Their Limitations

Inverse synthetic aperture radar (ISAR) imaging of the noncooperative maneuvering target is a challenging task because of its time variant orientation and rotation velocity which cannot be measured accurately. This correspondence investigates the principles of ISAR imaging of maneuvering targets, and proposes an algorithm for application in situations where the maneuverability is not too severe and the Doppler variation of subechoes from scatterers can be approximated as a first-order polynomial. The imaging results obtained by using real data show the effectiveness of the new method.

I. INTRODUCTION

Inverse synthetic aperture radar (ISAR) imaging has received much attention in the past two decades [1]. The principles of ISAR imaging of targets (e.g. aircraft) flying along a straight path with a constant velocity are basically the same as those of SAR. Compared with the scene illuminated by SAR, ISAR targets are much smaller in size, as a result the electromagnetic wave can be approximated to planar wave. With microwave radars, only a small variation in viewing angle, say 2–3°, is required to obtain an ISAR image with an adequate cross-range resolution. For small targets, the residual motion through range

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